

1. INTRODUCTION

Accurate geospatial data is essential to the architecture of many digital services, such as financial systems, e-commerce, and logistics [AddrLLM]. In Indonesia, the urgency for precise addressing is underscored by the explosive growth of the e-commerce sector, which saw an increase of nearly **500% between 2015 and 2019** [Evaluationof]. This rapid expansion has placed immense pressure on "last-mile" delivery providers to develop technology that ensures efficiency. However, Indonesian address data is marked by high variability and structural inconsistency [Issuesand]. Similar issues are observed across other developing countries, where the absence of a unified address system often leads to logistical bottlenecks. In the local context, some addresses are commonly written with random abbreviations (e.g., "Klp Gdng" for "Kelapa Gading"), local slang, or incomplete details, such as buildings referenced by kilometer markers rather than sequential house numbers [Issuesand].

This variability represents a "major challenge" because it results in **abnormal addresses** that fail to pinpoint a specific physical location. The impact potentially cost over **25,000 daily re-routing events**, leading to \$2 million annual losses [AddrLLM]. In Indonesia, even after the introduction of the national standard **SNI 9037:2021**, field observations in urban areas like Yogyakarta and Jakarta show that **only 3.1% of buildings currently comply** with standardized addressing rules. Common issues found at the site include one address being shared by 12 different buildings, non-sequential numbering, and the presence of "nested addresses" where users conflate two different regions in one line [Surveidan].

Traditional rule-based methods, such as regular expressions, tend to fail in capturing the semantic context needed to parse such noisy, unstructured text [FightingCrime]. Furthermore, standard address locators developed overseas (such as US-style ArcGIS systems) are often "not suitable" for Indonesia [Developing] because they fail to process unique community-based components like **Rukun Tetangga (RT) and Rukun Warga (RW)** [Issues]. While Large Language Models (LLMs) offer a promising solution due to their strong semantic understanding, their deployment for processing millions of daily packages costs massive computational overhead, high latency, and potential privacy concerns [SlimLM]. Additionally, LLMs are prone to **hallucinations**, where the model generates plausible but nonexistent address entities, which could lower the accuracy of the data used [FightingCrime].

This research focuses on Jakarta, a heavily populated metropolitan area that possesses a complex administrative structure with numerous neighborhood associations (RT/RW) and sub-districts (Kelurahan), often located within informal settlements or *kampungs*. The morphological irregularity of these areas, combined with subjective address naming by residents, creates a significant operational bottleneck, driving up delivery failure rates and operational expenses. These challenges highlight the necessity for effective solutions to

transform unstructured addresses into structured and extractable data that aligns with national standards.

To address these challenges, this study proposes a hybrid inference pipeline that combines a finely-tuned **Small Language Model (SLM)** with a rule-based validation system. SLMs, typically ranging from 125M to 8B parameters, have demonstrated a "sweet spot" between size, speed, and accuracy, making them ideal for on-device or resource-constrained deployment [SlimLM]. The SLM captures a variety of linguistic nuances and performs flexible parsing, while the deterministic rule-based component ensures structural consistency and prevents nonexistent outputs by validating them against **SNI 9037:2021** [Developing]. This approach supports a "Proof of Sufficiency" argument by showing that compact models can achieve industrial-grade performance without needing large-scale LLMs. Specifically, this work demonstrates that an 8B-parameter SLM can achieve competitive performance with significantly lower computational cost, introduces a deterministic validation mechanism to mitigate hallucination risks, and offers a scalable framework for transforming Jakarta's unstructured address text into standardized formats.

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2. PREVIOUS WORKS

2.1 Technical Evolution of Named Entity Recognition (NER) and Address Parsing

Named Entity Recognition (NER) is considered the "**cornerstone**" for information extraction, specifically for identifying location entities within unstructured text. **Basra Jehangir et al.** [ASurveyOn] highlight that NER is a crucial step for identifying appropriate entities to assist subsequent NLP activities. In a comprehensive review, **Wei Liang Seow et al.** [AReviewof] detail the methodology's transition from **rule-based strategies** to more flexible **statistical and neural models**, though they note that handcrafted rules often remain rigid and fail to generalize well.

Address parsing specifically extends the NER task by decomposing raw location strings into structured hierarchical components. **Beauchemin and Yassine** introduced the **Deepparse** project [Deepparse], an open-source library that leverages Seq2Seq bidirectional-LSTM architectures for multinational address parsing. In the context of Chinese addresses, **Liuchang Xu et al.** developed the **GSAM (Geospatial-Semantic Address Model)** [GSAM], which utilizes BERT to learn semantic representations fused with geospatial features. Similarly, **Singhal et al.** proposed the **GeoIndia** framework [GeoIndia], which addresses the lack of standardization in Indian addresses through hierarchical cell prediction.

2.2 Large Language Models and Geospatial RAG

The emergence of Large Language Models (LLMs) has introduced generative information extraction, enabling few-shot learning for complex geospatial tasks. **Qinchen Yang et al.** developed the **AddrLLM** framework [AddrLLM] that uses **Retrieval-Augmented Generation (RAG)** to rectify "abnormal" addresses by fetching external spatial context from nationwide logistics data.

Research by **Yingjie Hu et al.** on geo-knowledge-guided GPT models (such as ChatGPT and GPT-4) demonstrates that fusing location-description knowledge into prompts significantly improves extraction accuracy from social media [GeoKnowledge]. Furthermore, **Naveen et al.** introduced **GeoNLU** [GeoNLU], a framework designed to bridge the gap between natural language and spatial data infrastructures (SDIs) to enable seamless geospatial querying. Despite these capabilities, **Hammami et al** [FightingWith]. warn that generative LLMs are still prone to **hallucinations**, where they create nonexistent address tags not present in the original input.

2.3 Efficiency through SLMs and Knowledge Distillation

2.4 Indonesian Geospatial Standards and Challenges

Google Colab Link

- Proof of Concept Thesis:  [Google Colab](#)
- Generate Data:  [Google Colab](#)

Dataset

- <https://kodepos.co.id/kode-pos>
- <https://jdih.jakarta.go.id/dokumen/detail/10793/peraturan-gubernur-nomor-261-tahun-2015-tentang-penetapan-nama-jalan-pada-masing-masing-klas-jalan-sebagai-dasar-penghitungan-pajak-reklame>